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3-2 Project Based Learning in MACHINE LEARNING

Comparative Analysis of Dimensionality Reduction Techniques for Predicting Survival on the Titanic

Under the esteemed guidance of

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DONE BY

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Abstract

Dimensionality reduction techniques play a crucial role in enhancing the efficiency and interpretability of machine learning models by transforming high-dimensional data into lower-dimensional representations. Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) are two prominent methods used for this purpose. This study evaluates and compares the effectiveness of PCA and LDA in the context of the Titanic dataset, exploring their impact on model performance using classifiers like Logistic Regression, Random Forest, Decision Tree, and Support Vector Machine. Results indicate that while PCA simplifies data representation and aids in visualization, LDA often outperforms PCA in classification tasks, leveraging class-specific information to achieve higher accuracy.

INTRODUCTION

In this presentation, we explore the Titanic dataset to predict passenger survival using machine learning models. We compare the performance of models using Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) for dimensionality reduction. PCA is an unsupervised learning technique that identifies orthogonal components capturing maximum variance, making it ideal for data exploration and noise reduction. LDA is a supervised learning method focusing on finding linear combinations of features that best separate different classes, thereby improving classification accuracy. The main purpose of this project is to bridge the gap between theoretical concepts and practical application, demonstrating that LDA is best for supervised learning.

Dataset Overview

The Titanic dataset contains information on 891 passengers, including their Age, Sex, Fare, Embarked location, and survival status. The target variable is 'Survived', where 1 indicates survival and 0 indicates non-survival. The dataset requires preprocessing to handle missing values and encode categorical variables.

Data Preprocessing

Missing values in the Age column are filled with the median age, and missing values in the Embarked column are filled with the mode. Irrelevant columns such as PassengerId, Name, Ticket, and Cabin are dropped. Categorical variables (Sex, Embarked) are converted to dummy variables for model compatibility. The data is split into training (70%) and testing (30%) sets, and numerical features (Age, Fare) are scaled using StandardScaler.

Models and Parameter Grids

We use four machine learning models: Logistic Regression, Random Forest, Decision Tree, and Support Vector Classifier (SVC). Each model is tuned using GridSearchCV to find the best hyperparameters.

Logistic Regression: Hyperparameters include C and solver.

Random Forest: Hyperparameters include n_estimators, max_depth, and criterion.

Decision Tree: Hyperparameters include max_depth, min_samples_split, min_samples_leaf, and criterion. SVC: Hyperparameters include C, gamma, and kernel.

Principal Component Analysis (PCA)

PCA is a technique used to reduce the dimensionality of the dataset while retaining most of the variance. In this analysis, PCA reduces the dataset to two principal components. We visualize the decision boundaries of each model trained on the PCA-transformed data.

Linear Discriminant Analysis (LDA)

LDA is a technique used to reduce the dimensionality of the dataset while maximizing class separability. In this analysis, LDA reduces the dataset to one discriminant component. We visualize the decision boundaries of each model trained on the LDA-transformed data.

Model Evaluation - PCA

Logistic Regression (PCA): Best Params: {'C': 1, 'solver': 'liblinear'}, Accuracy: 0.6493
Random Forest (PCA): Best Params: {'n_estimators': 200, 'max_depth': 10, 'criterion': 'entropy'}, Accuracy: 0.6940
Decision Tree (PCA): Best Params: {'max_depth': 4, 'min_samples_split': 2, 'min_samples_leaf': 1, 'criterion': 'gini'}, Accuracy: 0.6604
SVC (PCA): Best Params: {'C': 10, 'gamma': 0.1, 'kernel': 'rbf'}, Accuracy: 0.6791
Random Forest achieved the highest accuracy with PCA.

Model Evaluation - LDA

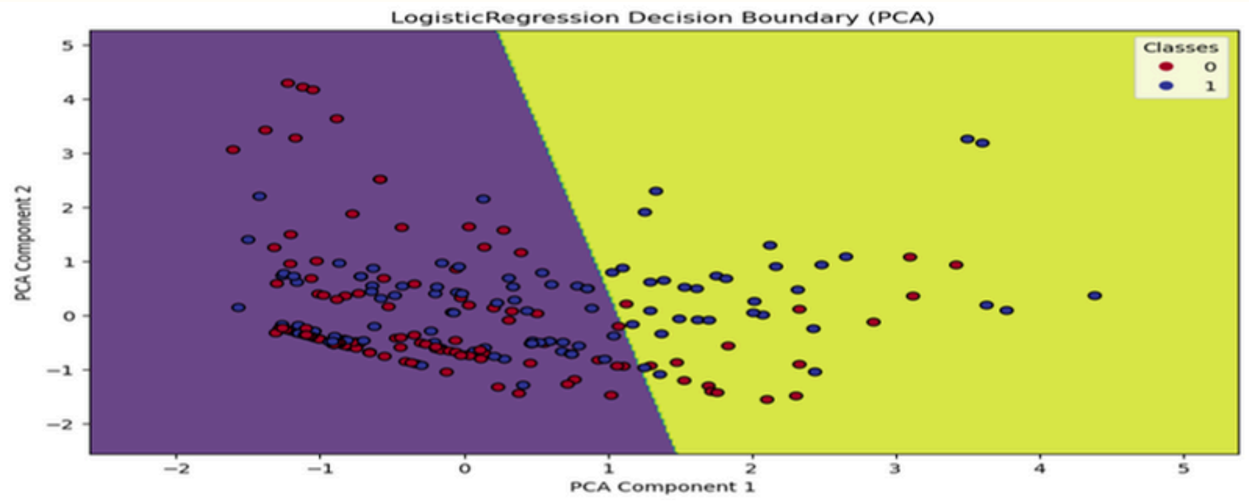
Logistic Regression (LDA): Best Params: {'C': 0.1, 'solver': 'liblinear'}, Accuracy: 0.8097
Random Forest (LDA): Best Params: {'n_estimators': 200, 'max_depth': None, 'criterion': 'entropy'}, Accuracy: 0.7799
Decision Tree (LDA): Best Params: {'max_depth': 10, 'min_samples_split': 2, 'min_samples_leaf': 1, 'criterion': 'gini'}, Accuracy: 0.7910
SVC (LDA): Best Params: {'C': 1, 'gamma': 1, 'kernel': 'rbf'}, Accuracy: 0.8097
Logistic Regression and SVC achieved the highest accuracy with LDA.

OUTPUT

PCA Results:

LogisticRegression Best Params (PCA): {'C': 0.1, 'solver': 'liblinear'}
LogisticRegression Accuracy (PCA): 0.6492537313432836

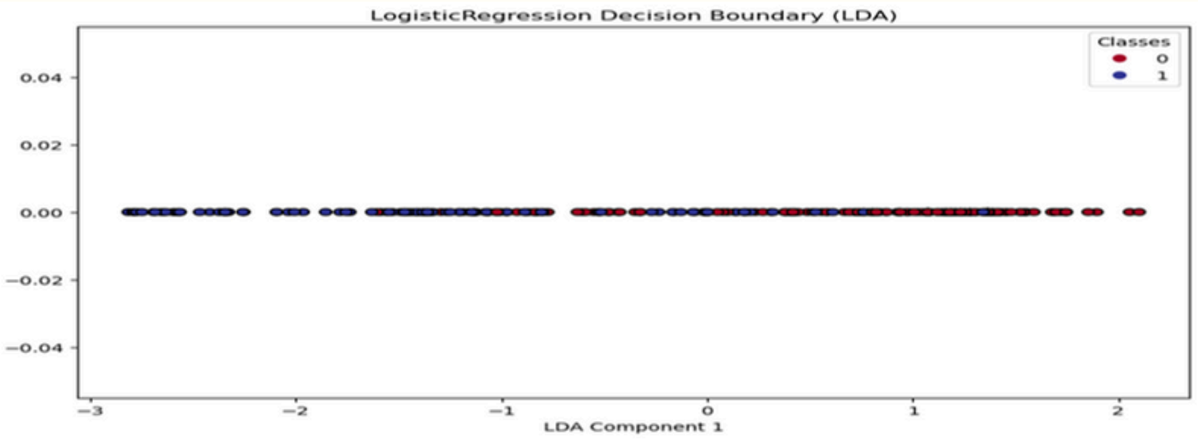
	precision	recall	f1-score	support
0	0.64	0.90	0.75	157
1	0.67	0.30	0.41	111
accuracy			0.65	268
macro avg	0.66	0.60	0.58	268
weighted avg	0.66	0.65	0.61	268



LDA Results:

LogisticRegression Best Params (LDA): {'C': 0.01, 'solver': 'saga'}
LogisticRegression Accuracy (LDA): 0.8097014925373134

	precision	recall	f1-score	support
0	0.80	0.90	0.85	157
1	0.83	0.68	0.75	111
accuracy			0.81	268
macro avg	0.81	0.79	0.80	268
weighted avg	0.81	0.81	0.81	268



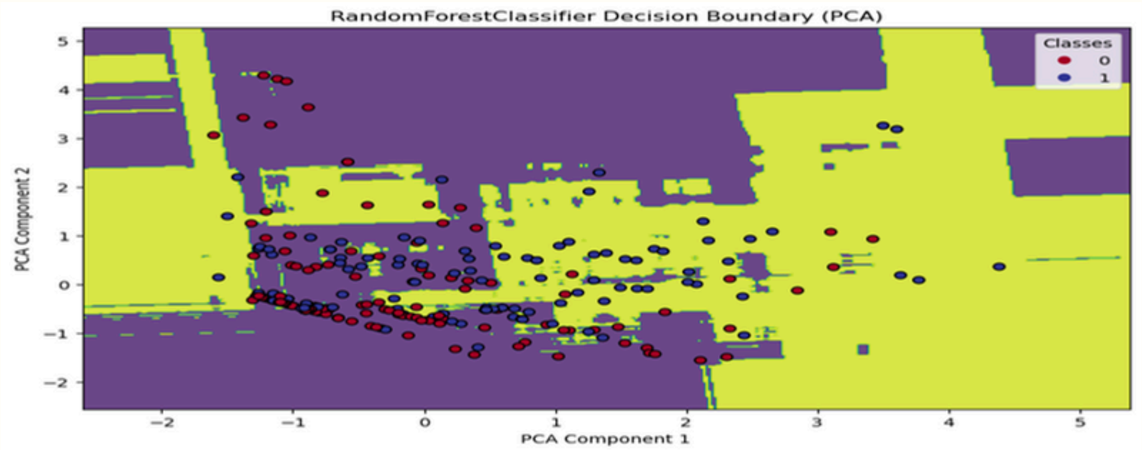
OUTPUT

PCA Results:

RandomForestClassifier Best Params (PCA): {'criterion': 'gini', 'max_depth': 10, 'n_estimators': 200}

RandomForestClassifier Accuracy (PCA): 0.6940298507462687

	precision	recall	f1-score	support
0	0.70	0.83	0.76	157
1	0.67	0.50	0.58	111
accuracy			0.69	268
macro avg	0.69	0.67	0.67	268
weighted avg	0.69	0.69	0.68	268

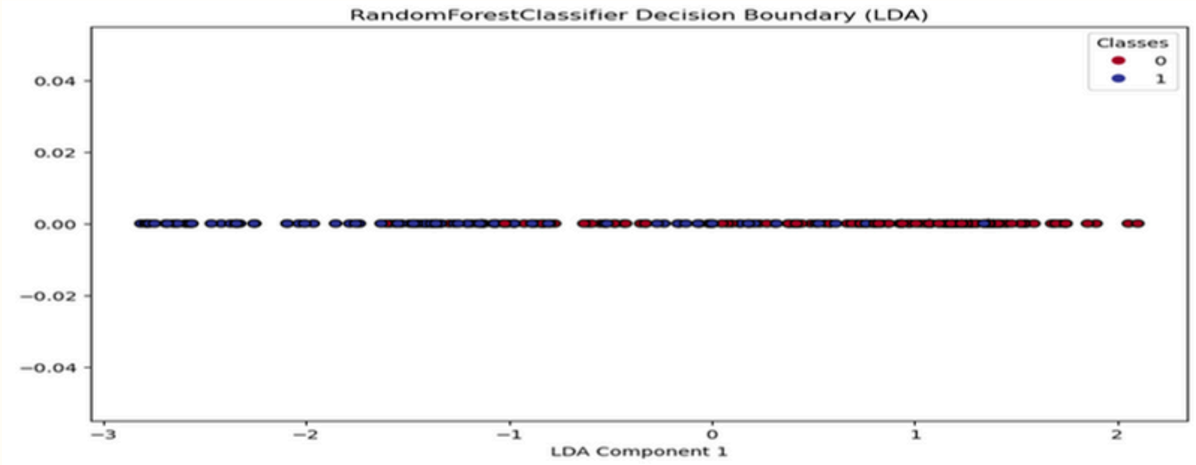


LDA Results:

RandomForestClassifier Best Params (LDA): {'criterion': 'entropy', 'max_depth': 10, 'n_estimators': 200}

RandomForestClassifier Accuracy (LDA): 0.7798507462686567

	precision	recall	f1-score	support
0	0.79	0.85	0.82	157
1	0.77	0.68	0.72	111
accuracy			0.78	268
macro avg	0.78	0.76	0.77	268
weighted avg	0.78	0.78	0.78	268

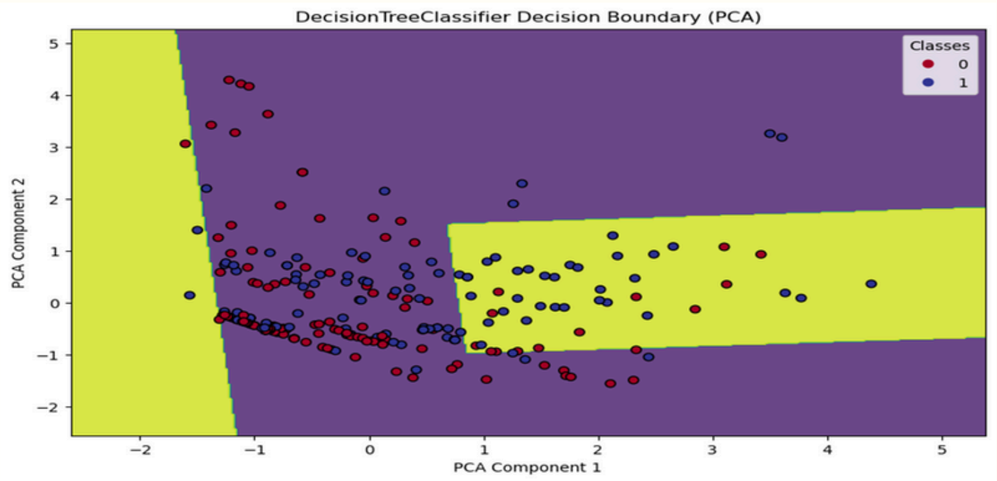


OUTPUT

PCA Results:

DecisionTreeClassifier Best Params (PCA): {'criterion': 'gini', 'max_depth': 4, 'min_samples_leaf': 4, 'min_samples_split': 2}
DecisionTreeClassifier Accuracy (PCA): 0.6604477611940298

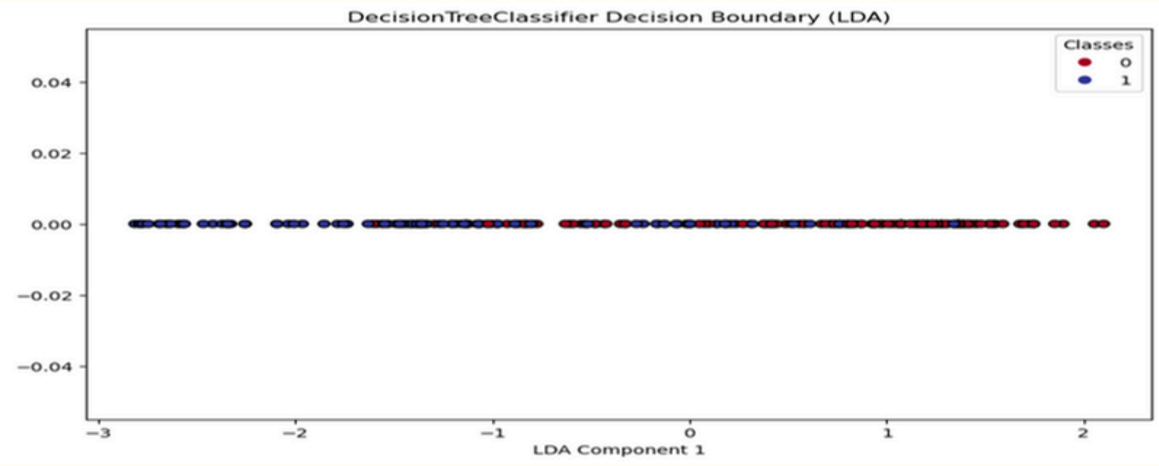
	precision	recall	f1-score	support
0	0.65	0.91	0.76	157
1	0.71	0.31	0.43	111
accuracy			0.66	268
macro avg	0.68	0.61	0.59	268
weighted avg	0.67	0.66	0.62	268



LDA Results:

DecisionTreeClassifier Best Params (LDA): {'criterion': 'gini', 'max_depth': 4, 'min_samples_leaf': 4, 'min_samples_split': 2}
DecisionTreeClassifier Accuracy (LDA): 0.7910447761194029

	precision	recall	f1-score	support
0	0.77	0.92	0.84	157
1	0.84	0.61	0.71	111
accuracy			0.79	268
macro avg	0.80	0.76	0.77	268
weighted avg	0.80	0.79	0.78	268

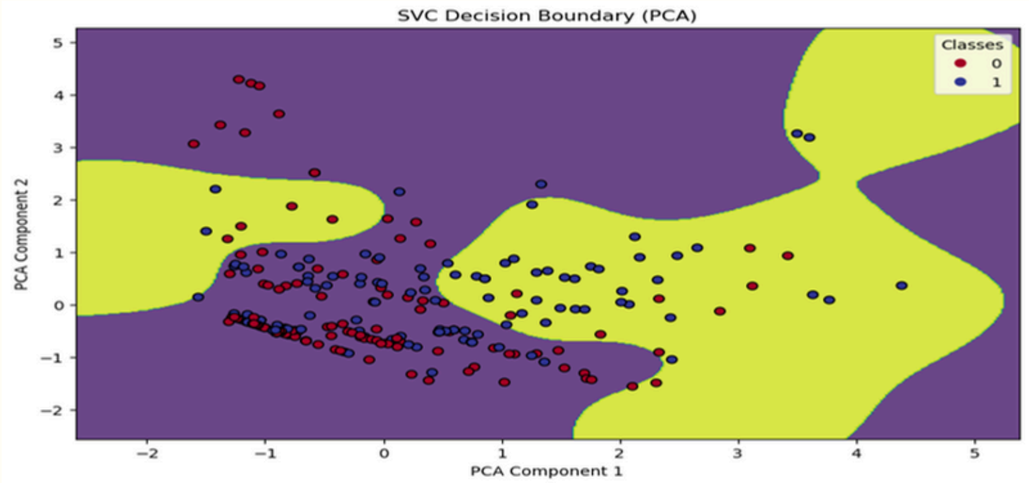


OUTPUT

PCA Results:

SVC Best Params (PCA): {'C': 10, 'gamma': 1, 'kernel': 'rbf'}
SVC Accuracy (PCA): 0.6791044776119403

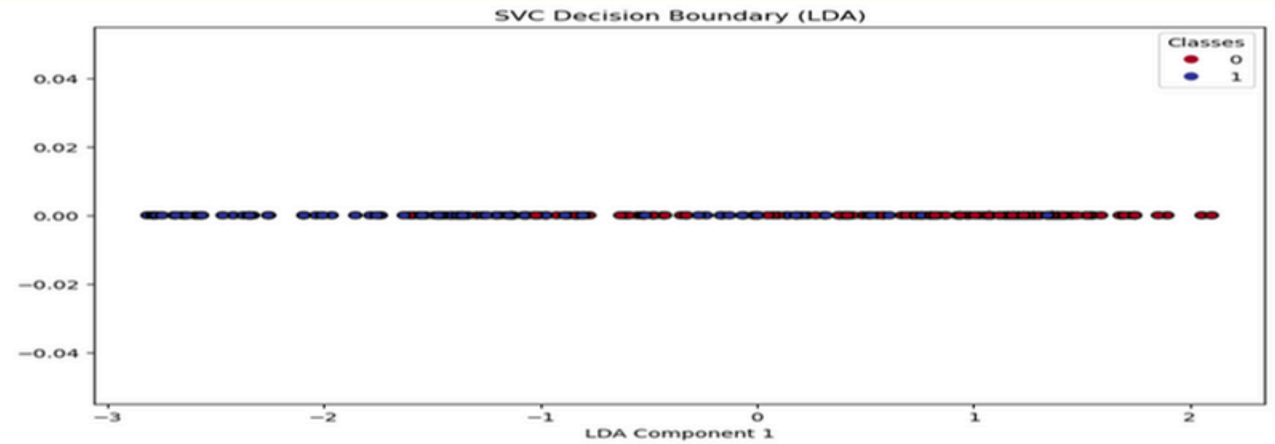
	precision	recall	f1-score	support
0	0.67	0.91	0.77	157
1	0.74	0.35	0.48	111
accuracy			0.68	268
macro avg	0.70	0.63	0.62	268
weighted avg	0.69	0.68	0.65	268



LDA Results:

SVC Best Params (LDA): {'C': 10, 'gamma': 1, 'kernel': 'rbf'}
SVC Accuracy (LDA): 0.8097014925373134

	precision	recall	f1-score	support
0	0.80	0.90	0.85	157
1	0.83	0.68	0.75	111
accuracy			0.81	268
macro avg	0.81	0.79	0.80	268
weighted avg	0.81	0.81	0.81	268



Summary of Accuracies

PCA Results: Logistic Regression: 0.6493, Random Forest: 0.6940, Decision Tree: 0.6604, SVC: 0.6791

LDA Results: Logistic Regression: 0.8097, Random Forest: 0.7799, Decision Tree: 0.7910, SVC: 0.8097

Random Forest performed best with PCA (0.6940).
Logistic Regression and SVC performed best with LDA (0.8097).

These results show that LDA, being a supervised method, generally provides better classification accuracy compared to PCA.

Best Model

The best model using PCA is Random Forest with an accuracy of 0.6940.
The best model using LDA is Logistic Regression with an accuracy of 0.8097.
Overall, LDA models achieved higher accuracy compared to PCA models, demonstrating the effectiveness of LDA for supervised learning tasks.

Conclusion

Random Forest achieved the highest accuracy with PCA, while Logistic Regression and SVC achieved the highest accuracy with LDA.

Dimensionality reduction techniques like PCA and LDA can help improve model performance and provide insights into the data.

The study confirms that LDA, a supervised learning technique, outperforms PCA in classification tasks by leveraging class-specific information.

Future work could explore other models, feature engineering, and cross-validation techniques to further improve accuracy.

In summary, while both PCA and LDA are powerful tools for dimensionality reduction, their suitability depends on the specific requirements of the task at hand. This study provides a comprehensive comparison, helping practitioners choose the most appropriate technique based on their needs.

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"A Gentle Introduction to Principal Component Analysis." Machine Learning Mastery, 2019. Online Article
"Linear Discriminant Analysis for Machine Learning." Towards Data Science, 2020. Online Article
2. Datasets:
Titanic Dataset from Kaggle: Kaggle Titanic: Machine Learning from Disaster
UCI Machine Learning Repository: Iris Data Set

Future Scope

The field of dimensionality reduction continues to evolve, with ongoing research aimed at addressing the limitations of existing techniques and exploring new methodologies. The following areas represent potential directions for future research and application:

Future directions for this project include:

Exploring Other Techniques: Compare PCA and LDA with methods like t-SNE and UMAP.

Advanced Feature Engineering: Implement advanced techniques to create new features.

Ensemble Methods: Evaluate ensemble learning methods for improved accuracy.

Deep Learning Integration: Assess deep learning models with dimensionality-reduced data.

Real-time and Streaming Applications: Extend analysis to dynamic environments.

Questions

Feel free to ask any questions.

Thank you for your attention!

THANK YOU FOR LISTENING!

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